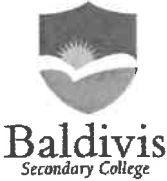


Name:	Markin Key	Date: 6/3/21
	Year 9 Mathematics Term 1 2021 Test 1: Algebra and Indices Time allowed: 50 minutes No calculators allowed Resources Allowed: • 1 piece of A4 paper notes • Pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters • Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.	

Total Marks:

/34

%

Question 1

[6 marks: 1, 1, 2, 2]

Simplify:

a) $7b - 8a + 5b + 10a$

(c)

$$7b + 5b - 8a + 10a$$

$$= 12b + 2a$$

b) $15p - 5p$

(a)

$$10p$$

c) $-3a + 2a - 5a$

(b)

$$-6a$$

d) $6m \times 9n$

$$54mn$$

e) $-6m \times -3m^3$

$18m^4$

f) $\frac{9m.m.m.m}{3m.m}$

$3m^2$ or $3m.m$

Question 2

[6 marks: 1, 1, 2, 2]

Evaluate the following:

a) $4 + 7 - 5$

$= 6$

c) $3 \times (1 + 2)^2 - 7$

$3 \times (3)^2 - 7$

$3 \times 9 - 7$

$27 - 7 = 20$

b) $7 \times (5 - (2 + 3)) + 8$

$7 \times (5 - 5) + 8$

$7 \times 0 + 8 = 8$

d) $(4)^2 - 8 - 7 + (-4 \times -8)$

$4 \times 4 - 8 - 7 + (-4 \times -8)$

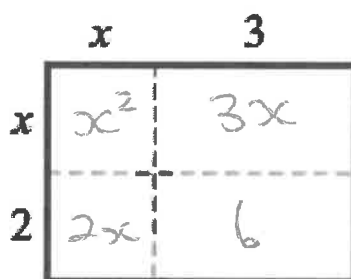
$16 - 8 - 7 + 32$

$= 33$

Question 4

[2 marks]

The diagram below shows the dimensions of a rectangle in terms of the variable x .



This area of this rectangle can be:

$(x + 2)(x + 3) = \underline{x^2} + 3x + \underline{2x} + \underline{6}$

$= \underline{x^2} + \underline{5x} + \underline{6}$

Question 5**[6 marks: $\frac{1}{2}, 2, 2$]**

Expand and simplify the following by collecting like terms and simplifying where possible.

a) $4(m - 5) + 5m$

c) $(t - 16)(t + 1)$

$$t^2 - 16t + t - 16$$

$$t^2 - 15t - 16$$

b) $2(l - 7) - 4(l + 3m)$

$$2l - 14 - 4l - 12m$$

$$2l - 4l - 14 - 12m$$

$$-2l - 14 - 12m$$

Question 6**[9 marks: 2, 2, 2, 3]**

Factorise the following expressions and simplify fully:

a) $12y^2 + 6y$

$$12x + 6$$

d) $6a + 2 + 3ab + b$

$$3a(2 + b) + b + 2$$

$$(3a + 1)(2 + b)$$

* c) $m(y + 2) + n(y + 2)$

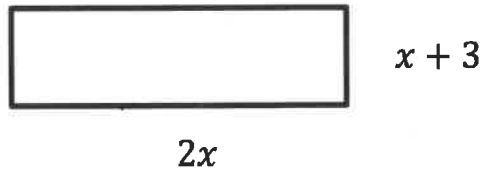
$$my + 2m + ny + 2n$$

$$(m + n)(y + 2)$$

e) $\frac{5a^2 - 5}{15a^2 - 15}$

Question 7**[3 marks: 1, 2]**

A rectangular field is $2x$ m long and $(x + 3)$ m wide.



- a) Find the expanded expression for the **area** of the field in terms of x .

x	x	$+3$
$2x$	$2x^2$	$+6x$

$2x^2 + 6x$

- b) The length of the fence surrounding the field can be calculated using the **perimeter** of the rectangular field. Give an expression for the perimeter of the field in terms of x , simplifying terms where possible.

$$2x + 2x + x + 3 + x + 3$$

$$4x + x + 6 + x$$

$$6x + 6$$